

Advanced Problems in Statistical Theory and Data Science

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This document presents a series of graduate-level problems in probability, statistical inference, data science, and machine learning. Each problem emphasizes theoretical derivation, interpretation, and numerical analysis.

PROBLEM 1. Zero-Inflated Normal Distribution

Let X be a zero-inflated normal:

$$\mathbb{P}(X = 0) = \alpha, \quad X \mid (X \neq 0) \sim \mathcal{N}(\mu, \sigma^2)$$

with probability $(1 - \alpha)$.

- (1) Write the PMF/PDF and CDF of X .
- (2) Derive $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (3) Numerically evaluate $\mathbb{P}(X \leq 0)$ for $\alpha = 0.3, \mu = 2, \sigma = 1$.

PROBLEM 2. Binomial Distribution Approximation

Let $X \sim \text{Bin}(n, p)$ with $n = 10, p = 0.3$.

- (1) Compute exactly $\mathbb{P}(X \geq 6)$.
- (2) Approximate $\mathbb{P}(X \geq 6)$ using the normal approximation with continuity correction and compare.

PROBLEM 3. Poisson Thinning

Suppose $N \sim \text{Poisson}(\lambda)$. Conditional on N , each of the N events is independently tagged “A” with probability q (and “B” otherwise). Let K be the number of “A” events.

- (1) Show $K \sim \text{Poisson}(\lambda q)$ (the thinning property).
- (2) Show that $K \mid N = n \sim \text{Bin}(n, q)$.
- (3) For $\lambda = 12, q = 0.2$, compute $\mathbb{P}(K = 0)$ and $\mathbb{P}(K \leq 2)$.

PROBLEM 4. Normalization and Scaling Transformations

Let $X = (X_1, \dots, X_n)$ be nonconstant real data with

$$\mu = \frac{1}{n} \sum_i X_i, \quad \sigma = \sqrt{\frac{1}{n} \sum_i (X_i - \mu)^2}.$$

Define:

- Z-score: $Z_i = \frac{X_i - \mu}{\sigma}$
- Min-max normalization: $M_i = \frac{X_i - \min X}{\max X - \min X}$
- Robust scaling: $R_i = \frac{X_i - \text{med}(X)}{\text{IQR}(X)}$

- Prove $\bar{Z} = 0$ and $\text{Var}(Z) = 1$.
- Show that Pearson correlation is invariant to positive affine transforms.
- Derive inverse transforms to recover X from Z and M .
- Perform numerical check for $x = (3, 7, 7, 19, 21)$.

PROBLEM 5. Histogram Rules and Robustness

You collect $n = 200$ observations: 199 are approximately $N(0, 1)$ and one outlier at 8. Compare three binning rules:

Sturges: $K = \lceil \log_2 n \rceil + 1$, Scott: $h_S = 3.5sn^{-1/3}$, FD: $h_{FD} = 2 \text{IQR} n^{-1/3}$.

- Compute the new mean and variance s^2 .
- Compute h_S and h_{FD} (use $\text{IQR} \approx 1.349$).
- On range $[-4, 8]$, report effective bin counts and interpret robustness.

PROBLEM 6. Logistic Regression on 2×2 Table

Two groups $X \in \{0, 1\}$ with outcomes $Y \in \{0, 1\}$:

	$Y = 1$	$Y = 0$
$X = 0$	a	b
$X = 1$	c	d

Model: $\text{logit}[P(Y = 1|X)] = \beta_0 + \beta_1 X$.

- Show $\hat{\beta}_0 = \log \frac{a}{b}$, $\hat{\beta}_1 = \log \frac{cb}{ad}$.
- Show $\text{Var}(\hat{\beta}_1) \approx \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$.
- Compute 95% CI for odds ratio using $a = 18, b = 12, c = 30, d = 10$.

PROBLEM 7. Within-Subjects ANOVA

Five subjects each measured under three conditions (within-subject). Data:

Subject	C1	C2	C3
1	10	12	13
2	9	11	12
3	8	10	11
4	9	10	12
5	11	12	14

- (1) Conduct a within-subjects ANOVA test for the condition effect (assume sphericity).
- (2) Compute the sums of squares, degrees of freedom, mean squares, and F -statistic.
- (3) Report whether the condition effect is statistically significant at $\alpha = 0.05$.

PROBLEM 8. Entropy, Gini, and Error Functions

For binary class probability $p \in [0, 1]$, define

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p), \quad G(p) = 2p(1 - p), \quad E(p) = \min\{p, 1 - p\}.$$

- (a) Prove $H(p) \geq G(p)$ for all p .
- (b) Prove $G(p) \geq E(p)$ for all p .
- (c) Provide a numerical comparison table for representative values $p = 0.1, 0.25, 0.5, 0.75, 0.9$.

PROBLEM 9. Information Gain and Attribute Selection

A dataset of 12 instances has $Y \in \{0, 1\}$ (6 positives, 6 negatives). Attributes:

- Attribute A: $v_1 : (5, 4+, 1-)$, $v_2 : (3, 1+, 2-)$, $v_3 : (4, 1+, 3-)$
- Attribute B: $b_0 : (6, 4+, 2-)$, $b_1 : (6, 2+, 4-)$

- (1) Compute parent entropy $H(S)$ and Gini $G(S)$.
- (2) Compute weighted entropy, information gain $IG(S; A)$, weighted Gini, and Gini decrease for A.
- (3) Repeat (2) for B.
- (4) Determine which attribute ID3 and CART would choose.

PROBLEM 10. Simple Linear Regression Analysis

Data ($n = 5$):

x	0	1	2	3	4
y	1.0	2.9	5.1	5.9	9.2

Model $y = \beta_0 + \beta_1 x + \epsilon$.

- (1) Compute $\hat{\beta}_0, \hat{\beta}_1$.
- (2) Compute residuals, SSE, R^2 , and residual standard error.
- (3) Test $H_0 : \beta_1 = 0$ (two-sided) and form a 95% CI for β_1 .
- (4) Compute a 95% prediction interval for y at $x_0 = 5$.

PROBLEM 11. Two-Piece Normal Distribution

Let X follow a split normal:

$$f(x; \mu, \sigma_1, \sigma_2) = \begin{cases} A \exp\left(-\frac{(x-\mu)^2}{2\sigma_1^2}\right), & x < \mu, \\ A \exp\left(-\frac{(x-\mu)^2}{2\sigma_2^2}\right), & x \geq \mu, \end{cases}$$

where $A = \sqrt{\frac{2}{\pi}}(\sigma_1 + \sigma_2)^{-1}$.

- (1) Show $f(x)$ integrates to 1.
- (2) Derive $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (3) For $\mu = 0, \sigma_1 = 1, \sigma_2 = 2$, evaluate $\mathbb{P}(-1 < X < 1)$ numerically.

PROBLEM 12. Gamma and Inverse-Gamma Relationship

Let $X \sim \text{Gamma}(\alpha, \beta)$ (shape $\alpha > 0$, rate $\beta > 0$), and $Y = 1/X$.

- (1) Derive the PDF of Y (the inverse-gamma distribution).
- (2) Compute $\mathbb{E}[Y]$ and specify when it exists.
- (3) Find the mode of Y given that $\text{mode}(X) = (\alpha - 1)/\beta$ for $\alpha \geq 1$.
- (4) For $\alpha = 4, \beta = 2$, compute $\mathbb{P}(Y > 1)$.

PROBLEM 13. Estimator Comparison for Exponential Mean

Let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$. We estimate the mean $\theta = 1/\lambda$ using:

$$T_1 = \bar{X}_n, \quad T_2 = \frac{n}{n+1} \bar{X}_n.$$

- (1) Compute the bias, variance, and mean squared error (MSE) of both T_1 and T_2 .
- (2) Determine which estimator has the smaller MSE, and identify the dependence on n and λ .
- (3) Show that T_1 is a consistent estimator for θ .
- (4) For $n = 25$ and sample mean $\bar{x} = 4.2$, compute T_1 and T_2 .

PROBLEM 14. Gini Coefficient and Inequality Measures

The Gini coefficient for a continuous distribution with CDF $F(x)$ and mean μ is:

$$G = \frac{1}{2\mu} \iint_{-\infty}^{\infty} |x - y| dF(x) dF(y).$$

- (1) For the Pareto distribution with $F(x) = 1 - (x_m/x)^\alpha$ for $x \geq x_m > 0, \alpha > 1$, show that:

$$G = \frac{1}{2\alpha - 1}.$$

- (2) For a sample $\{x_1, \dots, x_n\}$, estimate

$$\hat{G} = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x}},$$

and compute $\hat{G}$ for $\{1, 3, 6, 10\}$.

- (3) Prove that for any nonnegative data, $0 \leq \hat{G} \leq 1$.

PROBLEM 15. Bayesian Linear Regression Posterior

Consider the Bayesian linear model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2).$$

Priors: $\beta_0 \sim \mathcal{N}(0, \tau_0^2)$, $\beta_1 \sim \mathcal{N}(0, \tau_1^2)$, with known variances.

- (1) Write down $p(\beta_0, \beta_1 \mid \mathbf{y})$ up to proportionality.
- (2) Show that $p(\beta_1 \mid \beta_0, \mathbf{y})$ is Normal; identify mean and variance.
- (3) Given data from Problem 10 with priors $\tau_0^2 = \tau_1^2 = 100, \sigma^2 = 0.5$, compute posterior mean for β_1 (approximate via OLS).

PROBLEM 16. F-score and Precision–Recall Relationship

The F1-score is defined as:

$$F_1 = \frac{2PR}{P + R},$$

where P is precision and R is recall.

- (1) Express F_1 in terms of confusion matrix counts a, b, c, d from Problem 6.
- (2) Given classifier 1 with $(P, R) = (0.9, 0.7)$ and classifier 2 with $(0.8, 0.8)$, compute F_1 for both.
- (3) Prove that for $P, R \in (0, 1]$, $F_1 \leq \frac{P+R}{2}$ with equality iff $P = R$.

PROBLEM 17. Correlated Bivariate Normal Example

Let X, Y have joint PDF proportional to:

$$f_{X,Y}(x, y) \propto \exp\left(-(x^2 + xy + y^2)\right), \quad x, y \in \mathbb{R}.$$

- (1) Determine the normalizing constant.
- (2) Find the marginals of X and Y ; assess independence.
- (3) Derive the conditional $X \mid Y = y$.
- (4) Compute $\text{Cov}(X, Y)$.

PROBLEM 18. Quantile–Quantile (QQ) Plot and Skewness Assessment

A QQ-plot assesses whether data come from a target distribution.

- (1) Outline steps for constructing a QQ-plot against normality for a sample $\{x_1, \dots, x_n\}$.
- (2) For $n = 100$, with sample quartiles $Q_1 = -0.7, Q_3 = 0.6$ and theoretical normal quartiles $z_{0.25} = -0.674, z_{0.75} = 0.674$, compute the robust skewness:

$$\frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1},$$

and interpret its sign.

- (3) Explain what an “S-shaped” QQ-plot indicates about tail heaviness relative to normality.

PROBLEM 19. Principal Component Analysis (PCA)

Consider performing PCA on a data matrix $\mathbf{X}$ with n observations and p centered variables.

- (1) Prove that the first principal component direction $\mathbf{v}_1$ is the eigenvector of the covariance matrix $\mathbf{S}$ corresponding to its largest eigenvalue.
- (2) Show that the principal components are uncorrelated with each other.
- (3) Given

$$\mathbf{S} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix},$$

find the principal components and the proportion of total variance explained by the first principal component.

PROBLEM 20. Kullback–Leibler (KL) Divergence

The KL divergence between two continuous densities $p(x)$ and $q(x)$ is:

$$D_{\text{KL}}(P\|Q) = \int_{-\infty}^{\infty} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx.$$

- (1) Compute $D_{\text{KL}}(P\|Q)$ when $P \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Q \sim \mathcal{N}(\mu_2, \sigma_2^2)$.
- (2) Show that $D_{\text{KL}}(P\|Q) \geq 0$ with equality iff $P = Q$ almost everywhere.
- (3) Let $Q = \mathcal{N}(0, 1)$. For a sample $x_1, \dots, x_n$, show that minimizing $D_{\text{KL}}(\hat{P}\|Q)$ (where $\hat{P}$ is empirical) equals maximizing the log-likelihood under Q .

PROBLEM 21. Mixture of Two Gaussian Distributions

Let

$$X \sim \alpha \mathcal{N}(\mu_1, \sigma_1^2) + (1 - \alpha) \mathcal{N}(\mu_2, \sigma_2^2), \quad 0 < \alpha < 1.$$

- (1) Write the explicit form of the PDF of X .
- (2) Derive $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (3) For $\alpha = 0.6, \mu_1 = 0, \sigma_1 = 1, \mu_2 = 3, \sigma_2 = 2$, numerically evaluate $\mathbb{P}(X > 2)$.

PROBLEM 22. Negative Binomial Distribution

Let $X \sim \text{NegBin}(r, p)$, the number of failures before the r -th success in independent Bernoulli(p) trials.

- (1) Show that

$$P(X = k) = \binom{k+r-1}{r-1} p^r (1-p)^k, \quad k = 0, 1, 2, \dots$$

- (2) Derive $\mathbb{E}[X]$ and $\text{Var}(X)$ using the MGF or otherwise.
- (3) For $r = 3, p = 0.4$, compute $\mathbb{P}(X \leq 4)$ exactly.

PROBLEM 23. Uniform Estimators for θ

Suppose $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, \theta)$. Define:

$$T_1 = 2\bar{X}, \quad T_2 = \frac{n+1}{n}X_{(n)}, \quad X_{(n)} = \max(X_1, \dots, X_n).$$

- (1) Show that T_1 is unbiased for θ . Compute its variance and MSE.
- (2) Show that T_2 is unbiased for θ . Compute its variance and MSE.
- (3) For $n = 5$ and data $\{3.1, 1.6, 4.2, 2.9, 3.8\}$, compute both estimates.

PROBLEM 24. Mahalanobis Distance and Affine Invariance

The Mahalanobis distance for $\mathbf{x} \in \mathbb{R}^p$ relative to mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ is:

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}.$$

- (1) Prove that $D_M(\mathbf{x})$ is invariant under any affine transformation $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$, with invertible $\mathbf{A}$.
- (2) For $\boldsymbol{\mu} = (0, 0)^\top$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, compute $D_M((1, 1)^\top)$.
- (3) Explain why D_M is superior to Euclidean distance for detecting multivariate outliers.

PROBLEM 25. Hosmer–Lemeshow Goodness-of-Fit Test

The Hosmer–Lemeshow statistic for assessing logistic regression fit is:

$$H = \sum_{g=1}^G \frac{(O_g - E_g)^2}{E_g(1 - E_g/n_g)},$$

where O_g = observed number of successes in group g , E_g = expected number, n_g = group size.

- (1) Under H_0 (model is correct), show that $H \approx \chi_{G-2}^2$.
- (2) For a model with $n = 400$ and $G = 10$ groups, with computed $H = 8.75$, perform the test at $\alpha = 0.05$. State your conclusion.

PROBLEM 26. Power of a Two-Sample t -Test

Two independent samples:

$$X_1, \dots, X_{n_x} \sim \mathcal{N}(\mu_x, \sigma^2), \quad Y_1, \dots, Y_{n_y} \sim \mathcal{N}(\mu_y, \sigma^2).$$

Test $H_0 : \mu_x = \mu_y$ using

$$T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{n_x} + \frac{1}{n_y}}}, \quad s_p^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}.$$

- (1) Derive the non-centrality parameter δ when $\mu_x - \mu_y = \Delta$.
- (2) Show that the power increases with Δ and sample sizes, but decreases with σ .
- (3) For $n_x = n_y = 25, \sigma = 2, \alpha = 0.05$, approximate the power to detect $\Delta = 1$ using the normal approximation to the noncentral t .

PROBLEM 27. Receiver Operating Characteristic (ROC) and AUC

An ROC curve plots TPR vs. FPR across thresholds.

- (1) Prove that the Area Under Curve (AUC) equals $\mathbb{P}(\text{score}_{\text{pos}} > \text{score}_{\text{neg}})$.
- (2) For 5 positive and 5 negative scores:

Positives: $\{0.8, 0.6, 0.9, 0.7, 0.55\}$, Negatives: $\{0.4, 0.3, 0.45, 0.5, 0.35\}$,

compute the AUC using the probability ranking method.

- (3) Interpret $\text{AUC} = 0.5$ in terms of classifier skill.

PROBLEM 28. K-Means Clustering Algorithm

K-means partitions n points into k clusters by minimizing:

$$J = \sum_{i=1}^k \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2.$$

- (1) Define the objective function formally and state both the assignment and update steps.
- (2) Prove that each step of K-means (assignment and centroid update) never increases J .
- (3) Given data $\{(1, 1), (1, 2), (5, 4), (6, 5)\}$ with initial centroids $(1, 1)$ and $(5, 4)$, perform one full iteration and compute new centroids.

PROBLEM 29. Coefficient of Determination in Multiple Regression

For multiple regression with intercept, $R^2 = 1 - \frac{SSE}{SST}$.

- (1) Prove $R^2 = \rho^2(y, \hat{y})$, the square of the correlation between observed and fitted values.
- (2) Show that adding any predictor cannot reduce R^2 .
- (3) For $n = 30, p = 4, SSE = 45.2, SST = 120.5$, compute R^2 and the adjusted R^2 .

PROBLEM 30. Softmax Function and Numerical Stability

The softmax function for $\mathbf{z} \in \mathbb{R}^K$:

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad i = 1, \dots, K.$$

- (1) Prove that adding a constant c to all elements of $\mathbf{z}$ leaves $\sigma(\mathbf{z})$ unchanged.
- (2) Derive the Jacobian:

$$J_{ij} = \frac{\partial \sigma_i}{\partial z_j} = \sigma_i(\delta_{ij} - \sigma_j).$$

- (3) For $\mathbf{z} = (2, 1, 0)^\top$, compute softmax probabilities; repeat for $\mathbf{z} = (102, 101, 100)^\top$. Discuss numerical stability.

PROBLEM 31. Cauchy Distribution and Its Properties

The Cauchy distribution with location x_0 and scale $\gamma > 0$ has PDF:

$$f(x; x_0, \gamma) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]}.$$

- (1) Show that the mean of X is undefined and the variance is infinite.
- (2) Prove that if $X, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, then $Z = X/Y \sim \text{Cauchy}(0, 1)$.
- (3) For $X \sim \text{Cauchy}(0, 1)$, compute exactly $\mathbb{P}(-1 \leq X \leq 1)$.

PROBLEM 32. Ridge Regression Estimator

In ridge regression, coefficients are estimated by minimizing:

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \}.$$

- (1) Derive the closed-form solution $\hat{\boldsymbol{\beta}}^{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$.
- (2) Show that the ridge estimator is a linear shrinkage transformation of the OLS estimator.
- (3) For simple regression without intercept, with data $\mathbf{x} = (1, 2, 3)^\top$, $\mathbf{y} = (1, 3, 2)^\top$, compute ridge estimates for $\lambda = 1$ and $\lambda = 10$. Compare with OLS.

PROBLEM 33. Wishart Distribution

If $\mathbf{X}_1, \dots, \mathbf{X}_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$, then

$$\mathbf{W} = \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^\top \sim W_p(n, \boldsymbol{\Sigma}),$$

the Wishart distribution with n degrees of freedom and scale $\boldsymbol{\Sigma}$.

- (1) Show $\mathbb{E}[\mathbf{W}] = n\boldsymbol{\Sigma}$.
- (2) For $p = 2$, derive $\text{Var}(W_{12})$ in terms of Σ_{12} .
- (3) If $\mathbf{W} \sim W_2(5, \mathbf{I}_2)$, compute $\mathbb{P}(W_{11} + W_{22} > 10)$.

PROBLEM 34. Stochastic Gradient Descent (SGD)

For objective $J(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n L(\boldsymbol{\theta}; \mathbf{x}_i, y_i)$, SGD updates are given by:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t; \mathbf{x}_{i_t}, y_{i_t}),$$

where i_t indexes a randomly selected sample.

- (1) Write the SGD update rule explicitly and contrast it with batch gradient descent.
- (2) Prove that if the learning rate η_t satisfies the Robbins–Monro conditions

$$\sum_{t=1}^{\infty} \eta_t = \infty, \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty,$$

then SGD converges to a local minimum for convex J .

- (3) For $L(\theta) = \theta^2$, $\theta_0 = 3$, and learning rate $\eta_t = 1/t$, simulate three iterations of SGD with stochastic gradients $-4, 2, -3$.

PROBLEM 35. Bradley–Terry Model for Pairwise Comparisons

The Bradley–Terry model defines:

$$P(i > j) = \frac{\pi_i}{\pi_i + \pi_j}, \quad \pi_i > 0,$$

where π_i represents the latent strength of object i .

- (1) Prove that the model is invariant under scaling $\pi_i \rightarrow c\pi_i$, $c > 0$.
- (2) Derive the log-likelihood function given binary outcomes y_{ij} where $y_{ij} = 1$ if $i > j$, and 0 otherwise:

$$\ell(\boldsymbol{\pi}) = \sum_{i < j} \left[y_{ij} \log \frac{\pi_i}{\pi_i + \pi_j} + (1 - y_{ij}) \log \frac{\pi_j}{\pi_i + \pi_j} \right].$$

- (3) For three objects with comparisons: $A \checkmark B$ (3), $B \checkmark A$ (1), $A \checkmark C$ (4), $C \checkmark A$ (0), $B \checkmark C$ (3), $C \checkmark B$ (1), compute the MLEs of π_A, π_B, π_C (either via iterative scaling or score equations).
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PROBLEM 36. Bootstrap Confidence Intervals for the Median

A sample of size $n = 20$ from a right-skewed distribution is observed:

$$x = \{4.1, 4.5, 4.7, 4.9, 5.0, 5.2, 5.3, 5.5, 5.7, 5.8, 6.0, 6.1, 6.2, 6.5, 6.6, 6.8, 7.0, 7.1, 7.3, 7.4\}.$$

- (1) Describe how to construct a bootstrap percentile confidence interval for the population median.
- (2) Implement 1000 bootstrap resamples (conceptually) and approximate the 95% CI given the ordered sample quantiles $Q_{0.025} = 5.0$, $Q_{0.975} = 7.1$.
- (3) Discuss how the bootstrap interval compares in coverage accuracy to a normal-theory CI for symmetric data.

PROBLEM 37. Heteroscedasticity and Weighted Least Squares

Consider a linear model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, with $\text{Var}(\epsilon_i) = \sigma^2 x_i^2$.

- (1) Show that OLS is unbiased but inefficient under this heteroscedastic structure.
- (2) Derive the Weighted Least Squares (WLS) estimator and its variance.
- (3) Using the dataset $(x_i, y_i) = \{(1, 2.2), (2, 3.5), (3, 6.9), (4, 8.1), (5, 9.8)\}$, compute both OLS and WLS estimates of β_1 .
- (4) Compare their standard errors and interpret.

PROBLEM 38. Bayesian Inference for a Proportion

Let $X \sim \text{Bin}(n, p)$ with $n = 50$ and prior $p \sim \text{Beta}(\alpha, \beta)$.

- (1) Derive the posterior $p(p \mid X = x)$ and identify its distribution family.
- (2) For prior $\text{Beta}(2, 2)$ and observed $x = 18$, compute the posterior mean and 95% credible interval.
- (3) Compare this Bayesian interval to the frequentist Wilson interval for the same data.
- (4) Discuss sensitivity of posterior inference to the prior choice.

PROBLEM 39. Kernel Density Estimation and Bandwidth Selection

Given data $x_1, \dots, x_n$ drawn from an unknown continuous density $f(x)$, define the kernel estimator

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

where $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$.

- (1) Show that $\int \hat{f}_h(x) dx = 1$.
- (2) Derive expressions for the bias and variance of $\hat{f}_h(x)$ at a fixed point.
- (3) Using the normal reference rule $h_{\text{NR}} = 1.06 s n^{-1/5}$, compute h_{NR} for a sample of 200 points with sample sd $s = 3.4$.
- (4) Discuss the impact of choosing h too small or too large on bias and variance.

PROBLEM 40. Time-Series Autocorrelation and the Durbin–Watson Test

A simple regression model $y_t = \beta_0 + \beta_1 x_t + u_t$ is fit to time-ordered data.

- (1) Define the sample autocorrelation of residuals $\hat{\rho}_1 = \frac{\sum_{t=2}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2}$.
- (2) Show that the Durbin–Watson statistic $D = 2(1 - \hat{\rho}_1)$ and interpret its range.
- (3) For residuals $e = (1.2, 0.8, -0.3, -0.5, -1.0)$, compute D and test for positive autocorrelation at $\alpha = 0.05$.
- (4) Explain when the test is unreliable (e.g., presence of lagged dependent variables).

PROBLEM 41. Cross-Validation and Bias–Variance Trade-off

In k -fold cross-validation for a predictive model, the dataset is split into k equal folds.

- (1) Derive an expression for the expected prediction error as a function of bias and variance components.
- (2) Explain why LOOCV (leave-one-out) tends to have low bias but high variance.
- (3) Given 10-fold CV errors for different regularization parameters $\lambda = \{0.01, 0.1, 1, 10\}$:

$$\text{MSE} = \{12.5, 9.2, 8.7, 11.4\},$$

identify the optimal λ and estimate the generalization error.

- (4) Discuss why the minimum-MSE model may still overfit when model selection uncertainty is ignored.

PROBLEM 42. Survival Analysis and Kaplan–Meier Estimator

The following data record survival times (in months) for patients after treatment, with 1 = event (death) and 0 = right-censored.

Time	2	4	5	8	10	12	15	18	20
Status	1	1	0	1	0	1	1	0	1

Table 1: Survival times and censoring indicators.

- (1) Compute the Kaplan–Meier survival estimate at $t = 10$ and $t = 20$.
- (2) Plot (conceptually) the stepwise survival curve and indicate censoring marks.
- (3) Derive Greenwood’s formula for the standard error of $\hat{S}(t)$.
- (4) Interpret the estimated median survival time.

PROBLEM 43. Expectation–Maximization (EM) Algorithm for Gaussian Mixtures

Given data generated from a two-component mixture:

$$f(x) = \pi \mathcal{N}(x; \mu_1, \sigma_1^2) + (1 - \pi) \mathcal{N}(x; \mu_2, \sigma_2^2).$$

- (1) Write the complete-data log-likelihood assuming latent class indicators $Z_i \in \{0, 1\}$.
- (2) Derive the E-step and M-step update formulas for $\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2$.

(3) Using the following toy dataset:

x_i	2.1	2.5	5.2	5.8	6.1
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Table 2: Five observed data points for EM initialization.

initialize $\pi^{(0)} = 0.5, \mu_1^{(0)} = 2.0, \mu_2^{(0)} = 6.0, \sigma_1^{2(0)} = \sigma_2^{2(0)} = 1.0$ and compute one full EM iteration numerically.

PROBLEM 44. High-Dimensional Inference and LASSO Path

Consider the linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $p > n$. The LASSO estimator minimizes

$$\hat{\boldsymbol{\beta}}(\lambda) = \arg \min_{\boldsymbol{\beta}} (\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1).$$

- (1) Explain geometrically why sparsity arises in the LASSO solution.
- (2) For the dataset below, compute solutions at $\lambda = 0.5, 1.0, 2.0$ (using coordinate-descent updates conceptually):

Obs	x_1	x_2	x_3	x_4	y
1	1	0	0	2	5.1
2	0	1	1	0	4.3
3	1	1	0	1	6.2
4	0	0	1	1	4.9

Table 3: Toy dataset for LASSO path illustration.

- (3) Sketch qualitatively the coefficient shrinkage path as λ increases.

PROBLEM 45. Causal Inference Using Propensity Scores

A randomized trial was infeasible, so treatment assignment followed observational patterns. The dataset is summarized as:

ID	Age	Income	Health_Score	Treatment (1/0)
1	45	62	80	1
2	38	55	70	0
3	60	90	85	1
4	50	75	77	0
5	33	40	68	0
6	55	82	84	1

Table 4: Covariates and treatment assignment.

- (1) Fit a logistic regression model $\hat{e}(x) = P(T = 1 \mid X = x)$ for propensity scores.
- (2) Compute inverse-probability weights $w_i = \frac{T_i}{\hat{e}(x_i)} + \frac{1-T_i}{1-\hat{e}(x_i)}$.
- (3) Explain how weighted averages estimate the Average Treatment Effect (ATE).
- (4) Discuss overlap and positivity assumptions for valid causal inference.

PROBLEM 46. Principal Components on a Real Dataset

The following standardized measurements were obtained for four variables across six observations:

Obs	X_1	X_2	X_3	X_4
1	1.2	0.9	1.1	0.8
2	0.8	0.6	1.0	0.7
3	-0.5	-0.6	-0.4	-0.3
4	-0.8	-1.0	-0.9	-1.2
5	0.3	0.4	0.2	0.5
6	-0.2	-0.1	0.0	-0.2

Table 5: Standardized feature matrix for PCA computation.

- (1) Compute the sample covariance matrix $\mathbf{S}$.
- (2) Determine eigenvalues and the proportion of variance explained by each principal component.
- (3) Project the observations onto the first principal component and interpret its loading vector.
- (4) Suggest dimensionality reduction if 95% variance retention is required.

PROBLEM 47. Model Validation on a Real Regression Dataset

The table below records study hours (x_1), attendance (x_2), and exam scores (y) for eight students.

Student	Study_Hours	Attendance	Exam_Score
1	4	6	68
2	6	8	80
3	2	5	54
4	8	9	88
5	5	7	74
6	7	8	84
7	3	4	60
8	9	10	92

Table 6: Student study dataset for regression validation.

- (1) Fit the linear model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$ by OLS.
- (2) Compute fitted values, residuals, and R^2 .
- (3) Perform 4-fold cross-validation and report mean test-MSE.
- (4) Plot residuals vs. fitted values (conceptually) and comment on linearity and variance assumptions.

PROBLEM 48. Multiple Testing and False Discovery Rate

You have performed 10 independent hypothesis tests and obtained p-values: 0.002, 0.010, 0.021, 0.045, 0.0

- (1) Apply the Benjamini–Hochberg (BH) procedure at level $\alpha = 0.05$. Which hypotheses are rejected? Show the steps.
- (2) Compute the Bonferroni-adjusted p-values and compare the number of discoveries to BH.
- (3) Discuss briefly (2–3 sentences) why BH controls FDR under independence but may fail under certain dependence structures.

PROBLEM 49. Gaussian Process Regression — Predictive Posterior

Consider a Gaussian process prior on a regression function $f : \mathbb{R} \rightarrow \mathbb{R}$ with mean zero and squared-exponential kernel

$$k(x, x') = \sigma_f^2 \exp \left(-\frac{(x - x')^2}{2\ell^2} \right).$$

Assume noisy observations $y_i = f(x_i) + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma_n^2)$.

- (1) Write the joint prior for training outputs and the predictive distribution for $f(x_*)$ at a new point x_* (mean and variance), in matrix form.

- (2) Given one training point $x_1 = 0, y_1 = 1$, with $\sigma_f^2 = 1, \ell = 1, \sigma_n^2 = 0.1$, compute the predictive mean and variance at $x_* = 0.5$ (numeric expressions are fine).
- (3) Explain how hyperparameters $(\sigma_f, \ell, \sigma_n)$ are typically chosen in practice (two common approaches).

PROBLEM 50. Spectral Clustering on a Small Graph

Let the similarity (adjacency) matrix for 4 nodes be

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

- (1) Form the unnormalized graph Laplacian $L = D - A$. Compute L explicitly.
- (2) Find the eigenvalues and eigenvectors of L (analytic reasoning or numeric values allowed). Identify the eigenvector(s) used to partition into two clusters.
- (3) Based on the Fiedler vector, give the two-cluster partition and briefly interpret the result.

PROBLEM 51. Bootstrap Confidence Intervals for Regression Coefficients

Consider the simple linear regression from Problem 10 (with the five data points): $x = (0, 1, 2, 3, 4)$, $y = (1.0, 2.9, 5.1, 5.9, 9.2)$.

- (1) Describe the nonparametric bootstrap procedure to obtain a 95% confidence interval for β_1 (the slope).
- (2) Perform a conceptual 1000-sample bootstrap (no need to compute by hand): explain how you would compute the percentile and the basic/bootstrap-t intervals and list advantages/limitations of each.
- (3) If the OLS slope estimate is $\hat{\beta}_1$ and the bootstrap sample of slopes has mean $\bar{\beta}^* = \hat{\beta}_1$ and standard error s^* , write the formula for a bootstrap-t 95% CI.

PROBLEM 52. Model Selection: AIC, BIC and Asymptotics

Suppose you fit three nested models to the same data with log-likelihoods and parameter counts as follows:

- Model M1: log-lik = -120.5, $k = 3$
- Model M2: log-lik = -118.0, $k = 5$

- Model M3: $\log\text{-lik} = -117.2$, $k = 8$
- (1) Compute AIC and BIC for each model (assume sample size $n = 200$). Which criterion selects which model?
 - (2) Discuss why AIC and BIC can prefer different models and their asymptotic behaviors (which is consistent for prediction vs. consistent model selection).
 - (3) For nested models, state (without proof) how a likelihood-ratio test complements AIC/BIC and when you would prefer it.
-

PROBLEM 53. Probability Mass and Density Functions (PMF / PDF)

This section contains advanced problems about PMFs, PDFs, their transformations, convolutions, order statistics, and inference.

- (1) (Mixed discrete–continuous distribution) Let X satisfy

$$\mathbb{P}(X = 0) = \alpha, \quad X \mid (X \neq 0) \sim \text{Exp}(\lambda), \quad 0 \leq \alpha < 1, \lambda > 0.$$
 - (a) Write the PMF/PDF and CDF of X explicitly (make the point-mass at zero clear).
 - (b) Derive expressions for $\mathbb{E}[X]$ and $\text{Var}(X)$ in terms of α, λ .
 - (c) Numerically evaluate $\mathbb{P}(X > 1)$, $\mathbb{E}[X]$ for $\alpha = 0.25, \lambda = 1$.
- (2) (Convolution and sums) Let $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$ be independent.
 - (a) Derive the PDF of $S = X + Y$ by convolution. Give the forms for the two cases $\lambda_1 \neq \lambda_2$ and $\lambda_1 = \lambda_2$.
 - (b) For $\lambda_1 = 1, \lambda_2 = 2$, evaluate the PDF of S and compute $\mathbb{P}(S > 1)$ (reduce to a closed-form expression before numerical evaluation).
 - (c) Generalize: if $X_i \stackrel{iid}{\sim} \text{Exp}(\lambda)$ for $i = 1, \dots, m$, identify the distribution of their sum and give its mean and variance.
- (3) (Order statistics and densities) Let $X_1, \dots, X_n$ be i.i.d. continuous with PDF f and CDF F .
 - (a) Write the joint PDF of the order statistics $X_{(1)} < \dots < X_{(n)}$ and derive the marginal PDF of the k -th order statistic $X_{(k)}$.
 - (b) For $X_i \stackrel{iid}{\sim} \text{Unif}(0, \theta)$, show that $\mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta$ and construct an unbiased estimator for θ based on the maximum.
 - (c) Discuss how the density of the sample minimum changes when the parent distribution has an atom (point-mass) at zero.

- (4) (Heavy-tail discrete distributions and normalization) Consider the discrete power-law (zeta) PMF

$$p(k) = C k^{-s}, \quad k = 1, 2, \dots, \quad s > 1.$$

- (a) Show that the normalizing constant is $C = 1/\zeta(s)$ where $\zeta(s) = \sum_{k \geq 1} k^{-s}$ is the Riemann zeta function.
 - (b) Determine for which values of s the expectation $\mathbb{E}[K]$ and variance $\text{Var}(K)$ are finite; express $\mathbb{E}[K]$ in terms of zeta functions when it exists.
 - (c) Suppose you observe a sample $k_1, \dots, k_n$ from this family. Discuss briefly (2–3 sentences) challenges for consistent parameter estimation of s and one practical estimator you might use.
- (5) (MLE and Fisher information for discrete pmf) Let $X_1, \dots, X_n$ be i.i.d. from a geometric distribution with support $\{1, 2, \dots\}$:

$$\mathbb{P}(X = k) = p(1 - p)^{k-1}, \quad 0 < p < 1.$$

- (a) Derive the log-likelihood and the maximum likelihood estimator $\hat{p}$.
 - (b) Compute the Fisher information $I(p)$ for a single observation and give an approximate standard error for $\hat{p}$.
 - (c) For the sample $n = 10$ with observed values summing to $\sum_i X_i = 30$, compute $\hat{p}$ and an approximate 95% Wald confidence interval for p .
- (6) (Characteristic functions and inversion) The Laplace (double-exponential) distribution has PDF

$$f(x; \mu, b) = \frac{1}{2b} e^{-|x - \mu|/b}, \quad b > 0.$$

- (a) Compute the characteristic function $\varphi(t) = \mathbb{E}[e^{itX}]$ of this distribution.
- (b) Use the inversion identity (state it) to argue why the characteristic function uniquely identifies the distribution, and show (formally) how the Laplace PDF can be recovered from its characteristic function in principle.
- (c) Briefly discuss one advantage of working with characteristic functions when studying sums of independent random variables.

PROBLEM 54. Normalization and Scaling (Advanced)

This section contains intermediate-to-advanced problems about data normalization and scaling methods (z-score, min–max, robust scaling, whitening, batch normalization) and their effects on downstream methods (PCA, clustering, distance-based learners).

- (1) (Properties and inversion)

- (a) Let $Z_i = (X_i - \mu)/\sigma$ be the z-score transform of a variable X . Show that the transform is invertible given μ, σ , and derive the inverse transform. Prove $\bar{Z} = 0$ and $\text{Var}(Z) = 1$ for a finite sample.
- (b) For min–max scaling $M_i = (X_i - \min X)/(\max X - \min X)$, discuss conditions when the transform is not invertible and propose a stable inversion strategy if the original min/max are known approximately (numerical stability discussion).

- (c) Define robust scaling using the median and IQR. Compare its breakdown point to z-score scaling and show one scenario where robust scaling is clearly preferable.
- (2) (Whitening and PCA)
- (a) Given data matrix $\mathbf{X}$ (centered) with sample covariance $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top\mathbf{X}$, show how to compute a whitening matrix W such that $W\mathbf{X}$ has identity covariance. Present two common choices (PCA/zero-phase and ZCA) and write formulas.
- (b) Prove that whitening is not unique and explain the geometric meaning of this non-uniqueness.
- (c) For $\mathbf{S} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$, compute a PCA-based whitening matrix (show eigen-decomposition steps) and verify the whitened covariance is identity (symbolic or numeric).
- (3) (Effect of scaling on distance-based learners)
- (a) Consider two features with ranges vastly different: Feature A in $[0, 1000]$, Feature B in $[0, 1]$. Give a short proof/argument why Euclidean-distance KNN with no scaling is dominated by Feature A. Provide a small numeric example with two sample points and a query point where scaling changes the nearest neighbour.
- (b) For K-means clustering, describe analytically how scaling columns to unit variance alters the within-cluster sum of squares objective and thus can change cluster assignments.
- (c) Propose a defensible default scaling strategy for KNN and K-means in mixed-scale datasets and justify it.
- (4) (Batch normalization — forward and backward views)
- (a) Define the batch normalization transform for a mini-batch $\{x_i\}_{i=1}^m$:
- $$\mu_B = \frac{1}{m} \sum_i x_i, \quad \sigma_B^2 = \frac{1}{m} \sum_i (x_i - \mu_B)^2, \quad \hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta.$$
- Provide the forward-pass steps and explain the role of γ, β and ϵ .
- (b) Derive (briefly) the gradient of the loss L w.r.t. the input x_i in terms of upstream gradient $\partial L / \partial y_i$ (outline the chain rule steps; a compact final formula is sufficient).
- (c) Discuss two failure modes or pitfalls when applying batchnorm (small batch sizes, training vs inference statistics), and suggest mitigations.
- (5) (Normalization and PCA / regression interplay)
- (a) Show with algebra why performing PCA on standardized variables (zero mean, unit variance) is equivalent to performing PCA on the correlation matrix rather than the covariance matrix, and discuss when this is desirable.

- (b) Consider linear regression with predictors on different scales. Explain how standardizing predictors affects (i) interpretation of coefficients, (ii) numerical conditioning of $\mathbf{X}^\top \mathbf{X}$, and (iii) the effect of L2 regularization (ridge). Provide short formulas indicating the role of scaling in each effect.
 - (c) Given the dataset $X_1 = (1, 10, 100)$, $X_2 = (0.1, 0.2, 0.4)$ (three observations), compute the sample covariance before and after standardization, and comment on the first principal component direction in both cases (numeric answers suffice).
- (6) (Practical normalization choices and evaluation)
- (a) Propose a small simulation (steps outlined) to compare the effect of z-score, min-max, and robust scaling on KNN classification accuracy when predictors include outliers. Describe the evaluation metric and how you would present results.
 - (b) Define the concept of "information leakage" when normalization parameters are computed on the full dataset prior to cross-validation. Explain the correct procedure to avoid leakage and quantify the bias it introduces qualitatively.
 - (c) For a high-dimensional dataset with many sparse categorical indicators (one-hot encoding), discuss the pros/cons of standard scaling (zero mean/unit variance) applied to those indicator columns and suggest an alternative preprocessing pipeline.

PROBLEM 55. One-Hot Encoding and Categorical Features (Advanced)

This section explores one-hot (dummy) encoding, high-cardinality categorical variables, encoding alternatives, effects on different models, and dataset-specific questions.

- (1) (Dummy variable trap and design matrix rank)
 - (a) Let a categorical variable C take values in $\{A, B, C\}$. Write the full one-hot design matrix (including an intercept) for observations with categories A, B, C, A . Show algebraically why including all three dummy columns plus an intercept makes the design matrix rank-deficient. Explain the standard remedy and show the full-rank design.
 - (b) Generalize the argument to a categorical variable with k levels and explain how many dummy columns to include to avoid collinearity with an intercept.
- (2) (Interpretation in generalized linear models)
 - (a) Consider fitting a logistic regression for a binary outcome Y using a 3-level categorical predictor C encoded with dummies D_1, D_2 and baseline level omitted. Write the model and show how to obtain the log-odds and odds ratio comparing level 2 to baseline.
 - (b) Dataset (counts by category and outcome):

oprule Category	$Y = 1$	$Y = 0$
Red	30	70
Blue	45	55
Green	10	40

Table 7: Counts by category and binary outcome (example).

Using the table above, compute (i) the sample log-odds for each category, (ii) the MLE intercept and coefficient for the Blue dummy when Red is the baseline (treat this as a grouped logistic by counts), and (iii) the predicted probability for an observation in Blue.

(3) (High-cardinality categorical variables and hashing / target encoding)

- (a) Suppose a feature G has 10,000 distinct categories. Explain why naive one-hot encoding is impractical. Describe the hashing trick and derive the expected number of collisions if hashing into m buckets (assume uniform hash).
- (b) Describe target (mean) encoding with smoothing: give the formula for the smoothed target encoding $\tilde{t}_j = \frac{n_j \bar{y}_j + k\mu}{n_j + k}$ for category j (define all symbols). Explain the bias-variance tradeoff and one cross-validation strategy to avoid target leakage when fitting such encodings.

(4) (Effects on tree-based vs linear models)

- (a) Give a concise argument why decision trees (CART) can natively split on categorical variables without one-hot encoding, and explain when one-hot encoding may still be useful for tree ensembles (e.g., interaction handling in boosted trees).
- (b) Dataset for computing information gain: a categorical feature with four levels and class counts:

oprule Level	Positives	Negatives
L1	12	3
L2	5	10
L3	8	8
L4	2	7

Table 8: Categorical levels and class counts for information-gain calculations.

Compute the parent entropy and the weighted entropy after splitting by this categorical feature; report the information gain. Show the steps.

(5) (One-hot, distances and embeddings)

- (a) Show that for one-hot encoding of a categorical variable with k levels (using orthogonal dummy columns), the Euclidean distance between two observations that differ only in that categorical variable equals $\sqrt{2}$ if their categories are different (assuming single categorical variable), and 0 if same. Provide a brief derivation.

- (b) Propose and describe one embedding method (e.g., supervised embedding via SVD of co-occurrence matrix or entity embeddings learned in a neural network) that yields continuous vectors for categories. Describe how you would evaluate whether the embedding improves KNN classification over one-hot.
- (6) (Practical pipeline, unseen categories and missingness)
- (a) Provide a detailed recipe (ordered steps) for preprocessing a dataset containing categorical variables (with missing values and potential unseen categories at prediction time) to be used with a linear model and with a tree ensemble. Include how to handle missing categories and unseen categories at predict time.
- (b) Advanced dataset (mixed numeric and categorical) for applied questions:

oprule ID	User_City	Product_Category	Price	Visits	Converted
1	Austin	Sports	49.99	3	0
2	Boston	Electronics	199.00	1	1
3	Austin	Home	25.50	5	0
4	Chicago	Sports	59.00	2	1
5	Denver	Fashion	79.99	4	0
6	Austin	Electronics	129.99	2	1
7	Boston	Fashion	39.99	1	0
8	El Paso	Home	18.00	6	0
9	Chicago	Sports	49.99	0	0
10	Unknown	Misc	9.99	2	0

Table 9: Sample ecommerce dataset with mixed categorical and numeric features.

Using the dataset above:

- (i) Show the one-hot design matrix (columns) for User_City and Product_Category using the common convention of omitting one level per categorical variable. List the resulting column names.
- (ii) If you use target (smoothed) encoding for User_City computed on the full dataset, compute the encoded values (use conversion rate as target; use smoothing parameter $k = 1$ and global mean μ computed from the table). Then explain why this approach would be biased and how to compute it correctly without leakage.
- (iii) For prediction time, suppose a new user from User_City = "San Francisco" appears (unseen). Describe three strategies to handle this unseen category and discuss tradeoffs.

PROBLEM 56. Pearson Correlation Coefficient and Related Analyses

This section contains intermediate-to-advanced problems about Pearson correlation, hypothesis testing, robust alternatives, partial correlations, correlation networks, and high-dimensional correlation estimation. A dataset is provided for applied subproblems.

(1) (Theory and geometry)

- (a) Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ be observed values with sample means $\bar{x}, \bar{y}$. Show that the sample Pearson correlation

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

equals the cosine of the angle between the centered vectors $\mathbf{x} - \bar{x}\mathbf{1}$ and $\mathbf{y} - \bar{y}\mathbf{1}$. Explain the implications for interpreting correlation as a similarity measure.

- (b) Show that correlation is invariant to adding a constant to either variable and to multiplying either variable by a positive constant. What is the effect of multiplying a variable by -1 ?
- (c) Prove that if $\tilde{x} = (x - \bar{x})/s_x$ and $\tilde{y} = (y - \bar{y})/s_y$ are standardized, then the slope from regressing $\tilde{y}$ on $\tilde{x}$ equals the Pearson correlation r_{xy} .

(2) (Statistical inference and Fisher transform)

- (a) Derive the t-test statistic for testing $H_0 : r = 0$ and show it has a Student's t_{n-2} distribution under normality assumptions. Provide the formula relating r and t .
- (b) State the Fisher z-transformation $z = \frac{1}{2} \log \frac{1+r}{1-r}$ and derive an approximate (asymptotic) 95% confidence interval for the population correlation ρ based on z and its standard error.
- (c) Numerical example: For an observed sample correlation $r = 0.45$ with $n = 20$, compute the t-statistic, two-sided p-value, and the 95% CI using Fisher z-transform. Report numerical answers to two decimal places.

(3) (Robustness, ranks and outliers)

- (a) Describe two robust alternatives to Pearson correlation (e.g., Spearman rank correlation and biweight midcorrelation). Give the formulas and discuss sensitivity to outliers.
- (b) Consider the paired samples $(1, 1), (2, 2), (3, 3), (4, 4), (100, 100)$. Compute the Pearson and Spearman correlations and explain the difference in light of the outlier.
- (c) Explain how a permutation test for correlation is performed and when it is preferable to parametric t-tests.

(4) (Applied dataset: compute, compare, and test)

oprule ID	X_1	X_2	X_3	X_4	X_5	X_6
1	10.0	9.5	5.1	30	-8.0	9.8
2	12.0	11.8	6.0	28	-9.1	11.6
3	9.1	8.8	4.7	35	-7.2	8.9
4	11.0	10.9	5.5	32	-8.5	10.7
5	13.4	13.0	6.7	31	-9.5	12.9
6	8.5	8.2	4.2	36	-6.9	8.0
7	10.2	9.9	5.0	33	-8.0	9.7
8	200.0	195.0	98.0	29	-160.0	193.0
9	7.3	7.0	3.7	39	-6.3	7.5
10	14.1	13.8	7.0	27	-10.0	13.7
11	11.2	10.8	5.6	34	-8.6	10.9
12	9.5	9.3	4.9	37	-7.8	9.4

Table 10: Sample dataset with one obvious outlier (row 8).

- (a) Compute the sample Pearson correlation matrix for $X_1, \dots, X_6$ using the full dataset (show the matrix). Identify the three largest absolute correlations and report their sample values.
 - (b) Test the significance of each reported correlation using the t-test from part (2) and adjust the p-values for multiple comparisons using both Bonferroni and Benjamini–Hochberg procedures; report which pairs remain significant under each correction (use $\alpha = 0.05$).
 - (c) Remove observation 8 (the outlier) and recompute the Pearson correlation matrix and the three top correlations. Compare numerically and interpret which correlations were driven by the outlier.
 - (d) Compute Spearman rank correlations on the full dataset and compare to Pearson; comment on robustness illustrated by these numbers.
- (5) (Partial correlation and conditioning)
- (a) Define the partial correlation between X and Y given Z , show how it can be computed via residuals from linear regressions, and provide the formula in terms of the inverse covariance (precision) matrix.
 - (b) Using the dataset above, compute the partial correlation between X_1 and X_6 controlling for X_4 . Compute it once with the full data and once without the outlier; comment on the change and what it implies about confounding.
 - (c) Suggest a permutation-based way to test whether the partial correlation is significantly different from zero, and outline the algorithmic steps.
- (6) (Correlation networks and high-dimensional considerations)
- (a) From the correlation matrix (after removing the outlier), construct an undirected correlation network by thresholding absolute correlation at 0.7. List the edges and compute the degree of each node. Which variable is most central by degree?

- (b) Discuss shrinkage (Ledoit–Wolf) estimation of the covariance/correlation matrix in the high-dimensional regime (p comparable to n). Explain why shrinkage helps and give the Ledoit–Wolf form for the shrinkage estimator.
- (c) For p large, explain how the graphical lasso (L1-penalized inverse covariance) can be used to estimate a sparse partial-correlation graph. Describe how to select the penalty parameter and how to validate the resulting graph structure.